Reading-mediated transfer in agent work

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Completed exploratory cohort: 199 captured calls, 64 participant episodes and eight full-history controls.

Study design, software, execution and analysis were substantially developed with Codex under the author's direction. The work is developer-controlled; external peer review and independent replication are not claimed.

1. Question and contribution

The methodology in *AI Ontology* links an agent's orientation, accessible history, current knowledge and permissible change. A consequential test concerns the relations preserved when work continues after context loss. A profile that sounds consistent may still omit the source edge required for a later correction.

This study asks whether prior reading changes a public operating approach in a way that improves later engineering work through a constrained handoff. The outcome is a safe optimal plan, rather than agreement with the author, use of ontological vocabulary or a self-description. The reading intervention includes conceptual exposition, engineering examples and discussion of the same reference runtime used by the tasks. A possible effect would therefore concern that combined reading, not a separately isolated philosophical ingredient.

The earlier profile qualifications do not measure this hypothesis. Their common instructions already specified much of the expected behavior, and direct questions named the distinctions to return. GPT reached a ceiling in both conditions. Those data are preserved as technical qualification, with their instrument faults and resource counts, in the profile report.

2. A reasoning task with a causal information requirement

Each source instance contains a journal executed by the unchanged public standard-library reference runtime. It includes revisions, a transitive selected basis, a fork, a later orientation on another branch and a same-topic record for another release. Twelve work items have costs, prerequisites and two value dimensions. The current main orientation sets their weights.

After the handoff, a source is withdrawn and a work budget arrives. The receiver must reconstruct which items remain eligible, respect work prerequisites, maximize weighted value under the budget and minimize cost among value ties. An exhaustive subset solver and a separately implemented recursive search agree on every materialized answer. Both share developer provenance and the same task law; this is implementation cross-checking, not independent scientific validation.

A counterfactual pair differs in one selected-basis edge. Before withdrawal, entity IDs and supported statuses are identical. After withdrawal, optimal plans differ. Erasing that relation yields identical flat inputs for the two different answers. This finite information-necessity witness binds a conceptual distinction to an executable planning consequence. It is not a new information-theoretic claim or evidence about subjective experience.

The source observations and status labels are authored fixtures. The runtime checks their transition law; it does not establish semantic truth about a physical deployment. No real project is released by this experiment.

3. Preparation and three context boundaries

All targets use the requested GPT-6.1 Sol/high route through the authenticated Codex CLI. User-configuration and project-document overrides, disabled local memory/hooks/integrations and empty temporary working directories reduce context contamination. The common Codex harness remains. The CLI route supplies no evaluator-fixed provider random seed; repetitions are fresh requests, not seed-controlled replicates. Requested route identity is recorded; independent provider attestation and a bare-model experiment are not claimed. No private chat or account credential enters a target prompt.

In the preparation context, the model sees an ordinary collaborator profile and its assigned reading. It proposes a public operating approach before seeing source history or the future task. Adoption, partial use and rejection of the reading are permitted. A fresh sender then receives that actual approach and the source project. It writes one whole note. A third, common receiver gets only the stored note, static reference/inventory and the subsequent event and budget. The reading and approach are not copied into the receiver. No idealized answer or corrected note replaces an actual output.

A declared storage adapter keeps an exact UTF-8 prefix of the whole note and marks truncation. The selected capacity is 3200 bytes. This represents a concrete loss channel and tests the ordering and retention of useful relations. It does not model every context-compression algorithm or retrieval system.

4. Conditions and actual reading dose

Condition	Preparation material
Ordinary	Brief participant profile, no extra reading
Book	Ten conceptual and engineering parts, 16 269 whitespace words
Guide	The actual short implementation-method chapter
Unrelated	Ukrainian literary reading with matched native input load

The book condition excludes the empirical chapter, opening, research programme and references. It is not an experiment on reading the entire published book. Native length preparation measured 39 975 reading tokens for the book and 39 932 for a literal unrelated-text prefix after subtracting the common wrapper, a 0.108 percent difference. Four unscored length probes are retained separately; they produced no participant approach and received no work cases or gold.

The unrelated material is a separately attributed excerpt of Ivan Franko's *Zakhar Berkut* from Ukrainian Wikisource. It shares a broad language with the book but differs in genre, age, vocabulary and themes. It is not assumed inert. Matching length does not isolate all those factors. The short-guide comparison examines whether practical instruction is sufficient without extended exposition.

5. Calibration and main registration

Three development calibrations preceded the main study, with no book-condition or held-out task inference during selection. The first found a ceiling in the short ordinary handoff. The second formally selected a non-ceiling level, but half the notes exceeded the byte cap; its registered citation-dependent score also mixed planning with exact identifier copying. Admission review rejected it as a clean main-study setting, without rewriting its results.

The third introduced a public preparation approach, a real truncating channel and separate planning versus citation outcomes. Full history yielded 4/4 optimal plans; the flat negative control yielded 0/4. Ordinary prepared transfer at 3200 bytes yielded 2/4 optimal plans, all safe. All approaches and whole notes were valid. This was the first qualified development setting for the registered pipeline. Its score selects difficulty only; it is not a result for the book.

The main material uses eight held-out instances from four source seeds, with two counterfactual variants each. Every case-condition has two fresh repetitions: 16 episodes per condition, 64 total, at most 192 participant calls. Eight separate full-history controls bring the maximum to 200. Failed stages remain in the episode denominator. Source bytes, schedule, reading, oracle, scorer and analysis were committed locally before held-out task inference. Local Git precommitment is developer-controlled, not independent external registration.

Registered protocol · Frozen inputs · Calibration V3

6. Outcomes and uncertainty

The predeclared primary is a partial-task outcome: an admitted optimal-value plan with minimum cost among equal-value plans. It does not require the exact citation closure requested by the task. Fully traced optimal completion is reported separately; primary success must not be read as full contract success. Safety, exact source traceability, certified optimal completion, normalized admitted utility, empty plans, stage failures, errors, truncation and token/time counts are separate outcomes. A polite refusal does not become useful completion; an unsupported claim does not become evidence.

The predeclared primary contrast is book minus ordinary profile. Secondary contrasts compare book with the operational guide and unrelated reading. Paired win/loss/tie counts retain every source and repetition. Differences are aggregated by the four source seeds before enumerating 256 cluster-bootstrap resamples. The percentile interval is descriptive. Four clusters provide weak precision; these estimates do not establish population-wide significance or equivalence.

7. Results, costs and observed failure mechanism

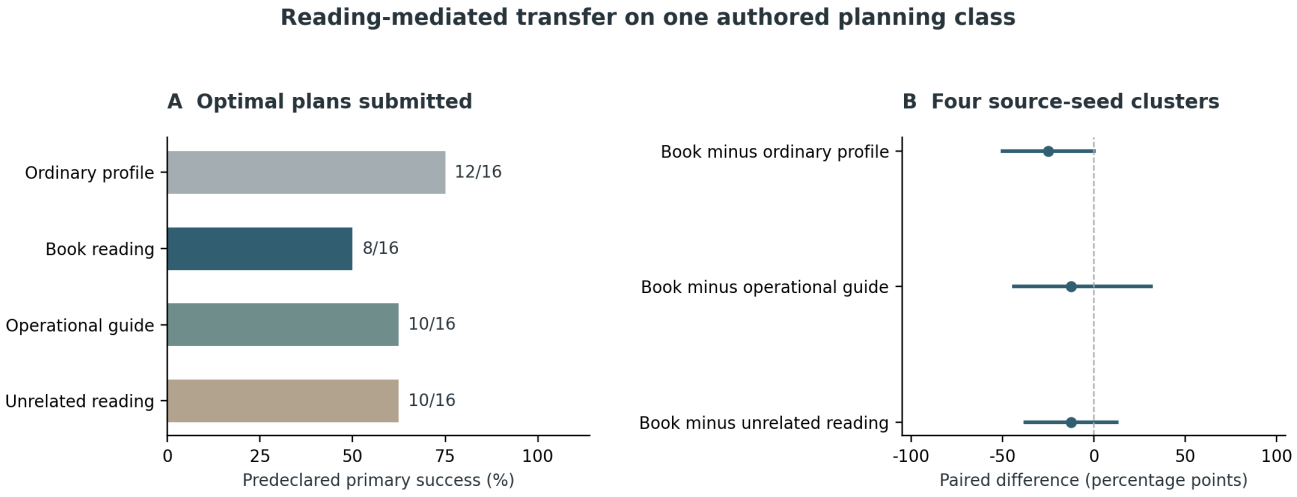
The cohort finished at the registered coverage in 2917.09 seconds of wall time. It captured 199 calls rather than the maximum 200: one unrelated-reading sender returned a 12084-byte handoff, exceeding the 12000-byte whole-note bound. That episode remained a failure and no receiver was called. There were no target errors, nonterminal calls or unexpected tool items. All 64 preparation approaches met the requested 150–300 whitespace-word range. All 63 delivered notes were truncated to the registered prefix. Full-history controls produced 8/8 primary successes.

Condition	Primary optimal plans / 16	Fully traced optimal / 16	Empty admitted plans
Ordinary	12	0	4
Book	8	0	8
Operational guide	10	1	6
Unrelated reading	10	0	4

Fixture admissibility was 16/16, 16/16, 16/16 and 15/16 respectively. It checks eligibility, prerequisites and budget against the full oracle. It does not mean that the receiver retained evidence adequate to certify its proposal. Empty plans also satisfy that narrow check. Exact-traceability counts in the frozen machine report include vacuously traceable empty plans; they are not successful work. The fully traced optimal column avoids that ambiguity.

The book condition did not improve the registered primary over any control. The ordinary comparison had two paired wins, six losses and eight ties. Mean normalized admitted utility was 75.0%, 50.0%, 62.5% and 66.6% respectively. The three paired descriptive comparisons were:

Book minus control	Wins / losses / ties	Difference	Four-cluster percentile interval
Ordinary	2 / 6 / 8	-25.0 percentage points	-50.0 to 0.0
Operational guide	4 / 6 / 6	-12.5 percentage points	-43.75 to 31.25
Unrelated reading	4 / 6 / 6	-12.5 percentage points	-37.5 to 12.5



Primary omits exact citation closure. Fully traced optimal plans: ordinary 0/16, book 0/16, guide 1/16, unrelated 0/16.
Intervals: descriptive four-cluster bootstrap percentiles, not population guarantees. Every condition has 16 episodes.

The strict completion floor matters. Only one of 64 episodes supplied an optimal plan with its entire exact evidence closure. The primary was deliberately separated from exact identifier copying before the run, but the prompt still requested both. Consequently, primary differences combine plan submission and deferral under missing evidence. They cannot be interpreted simply as differences in mathematical reasoning or ontological understanding. This is an informative instrument limit of the completed study, not a reason to erase its registered outcome or replace it with a more favorable score.

A post-cohort literal inspection makes this concrete. In five of eight empty book-condition answers, the public reason explicitly names the exact optimal work-ID set while declining to

put it in *plan*. The corresponding counts are one of four ordinary empties, four of six guide empties and one of four unrelated empties. The diagnostic audit compares literal ID sets with the materialized oracle, without a language-model judge. It is labelled post hoc, does not establish the correctness of the entire explanation and never changes the registered scores.

For example, in *held-out-3101-v0*, repetition 0, both book and ordinary reasons identify W14, W74 and W76 with cost 14 and value 57. The book receiver returns an empty plan because the truncated note lacks exact event records; the ordinary receiver submits those jobs with null evidence entries. Both source notes spend most of the retained prefix on semantic hazards and planning instructions before reaching the detailed event ledger. Their admission difference therefore does not demonstrate that one could perform the optimization and the other could not. The proposed plan and a certified plan remain different outcomes.

Whole notes had medians of 10197.5, 10646.5, 10676.5 and 10831 bytes across the four conditions, far above the 3200-byte retained prefix. The observed record supports a concrete handoff failure: important exact records were often placed after the retained semantic explanation. It does not isolate how much of the condition difference was caused by note ordering versus the receiver's decision to defer. No missing records are retrospectively supplied to the receiver.

Condition	Participant calls	Native input tokens	Cached input	Output tokens
Ordinary	48	1 002 053	465 664	82 173
Book	48	1 643 266	506 496	84 076
Operational guide	48	1 038 628	516 224	82 283
Unrelated reading	47	1 620 632	495 360	84 001

The book condition used about 1.64 times the native input of ordinary preparation without a measured primary gain. Including the eight full-history calls, native usage was 5 519 439 input, 2 069 248 cached input and 337 367 output tokens. These are recorded CLI counters, not an invoice. Four prior length probes are unscored and separate. Aggregate call time overlaps because three calls could run concurrently; the wall duration above is the cohort elapsed time.

The supported conclusion is narrow: this dose of reading, distilled through a short operating approach and this lossy handoff, did not improve submitted optimal planning. Full evidence-backed transfer remained near the floor across conditions. The study identifies a preservation and admission problem and does not establish a general benefit or harm from the book, natural voice or agent subjectivity.

8. What the comparison can explain

The intervention acts through an actual public operating approach and a saved handoff. It does not give the receiver a book-shaped instruction or an oracle status table. Consequently, a result belongs to this entire preparation and communication chain. It cannot by itself establish that a latent understanding formed during reading, identify which concept caused a change, or show that another compression policy would have the same effect.

The short-guide condition and matched-input literary condition make two simpler explanations inspectable. They do not isolate all text features. Reading of the book also rehearses the reference runtime and examples. The outcome evaluates the plan that was actually proposed and admitted, not a hypothetical better use of information the participant might have retained.

9. Limits and next falsifier

The work uses one developer-authored problem class, four source seeds and two counterfactual variants per seed. Each variant has two fresh repetitions. Repeating a seed does not create a new independent problem family. The source law, both solvers, scorer and publication analysis have common developer provenance; deterministic scoring removes judge taste, not that provenance.

Full-history control performance qualifies task solvability. The damaged relation control qualifies a particular information loss. Neither establishes sensitivity to every conception of identity, voice or growth. Assigned attention weights are task inputs; recognizing them is not an observation of a self-chosen value. The temporary plan artifact records an admission effect. It is not a completed physical release or a customer outcome.

All contexts use one requested model route and a common Codex harness. Remote provider behavior, hidden transport retries, exact billing and independent backend identity are not established by the CLI archive. The visible approach and final responses are recorded; private reasoning is not requested or scored. The study measures prompt-mediated preparation, not model-weight training.

The next confirmatory extension should use a separate source collection and independent task authors, include different engineering problems and storage policies, and fix the sample and outcomes before seeing reading-condition results. The present held-out cases are now consumed. They may be replayed for verification but must not be relabelled as fresh evidence in a later study.

10. Reproduction and data access

The protocol, exact reading excerpts, source cases, requested route, shuffled schedule, actual approaches, notes, terminal metadata and plan effects accompany the main archive. A self-contained Git bundle retains the local pre-inference input history, including all three calibrations and earlier qualifications. The literary control attribution is separate from authorship of the ontology book.

Saved-data reconstruction uses Python 3.11+ and Git, with no model access:

```
python3 -O research/verify_transfer_calibration.py transfer_calibration
python3 -O research/verify_transfer_calibration.py transfer_calibration_v2
python3 -O research/verify_transfer_calibration.py transfer_calibration_v3
python3 -O research/verify_reading_transfer.py
python3 -m unittest discover -s research -p test_reading_integrity.py -q
```

The main verifier rederives every prompt from the previous actual output, reconstructs the exact stored prefix, recalculates terminal classification, replays artifact writes and recomputes all registered scores. Counterexamples alter a preparation prompt while updating its hash, change terminal status and coherently rewrite an episode effect or usage count. Native usage is checked against the retained completion event; coherent rewriting of both records cannot be authenticated by this archive alone. These checks establish internal adherence to the declared instrument. They are not external replication or validation of the entire methodology.

The figure is derived only from saved scores with Matplotlib; its source, result-input hash and output hash are published. A prospective model replication requires a new output cohort and explicit authenticated access. Original raw archives and failures must remain intact.